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In [1]: import numpy as np
import matplotlib.pyplot as plt
import matplotlib.cm as cm

import warnings
warnings.filterwarnings(action='ignore')
```

Gradient

Berechnen Sie den Gradienten von

$$z(x, y) = 3x \ln\left(\sqrt{x^2 + y^2}\right) - \cos(xy^2 - \pi x)$$

am Punkt $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Lösungen

partielle Ableitungen:

$$z_x = 3 \ln(\sqrt{x^2 + y^2}) + \frac{3x^2}{x^2 + y^2} + (y^2 - \pi) \sin(xy^2 - \pi x)$$

$$z_y = \frac{3xy}{x^2 + y^2} + 2xy \sin(xy^2 - \pi x)$$

Gradient:

$$\nabla z = \begin{bmatrix} z_x \\ z_y \end{bmatrix} = \begin{bmatrix} 3 \ln(\sqrt{x^2 + y^2}) + \frac{3x^2}{x^2 + y^2} + (y^2 - \pi) \sin(xy^2 - \pi x) \\ \frac{3xy}{x^2 + y^2} + 2xy \sin(xy^2 - \pi x) \end{bmatrix}$$

Gradient am Punkt $P = (1, 0)$:

$$\nabla z = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$$

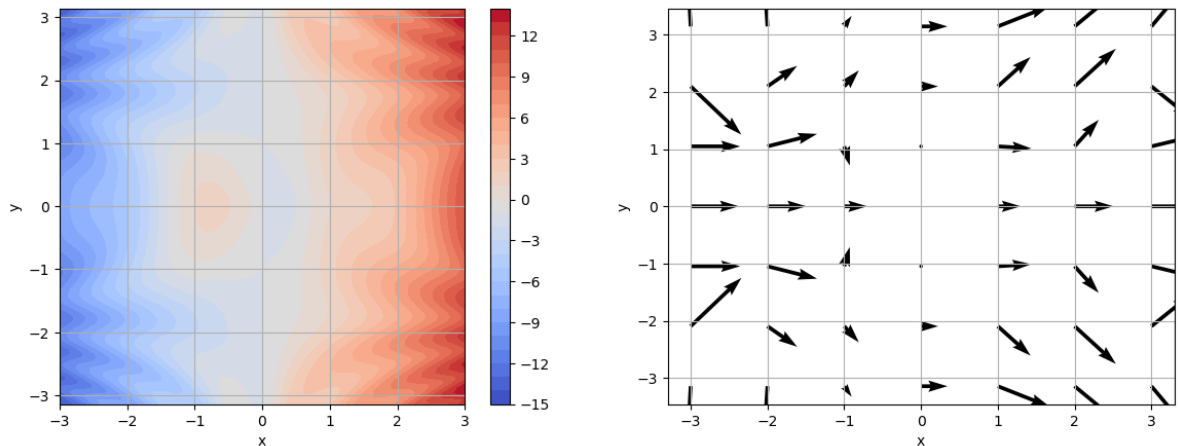
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In [2]: fig = plt.figure(figsize=(14,5))

x = np.linspace(-3,3,101)
y = np.linspace(-np.pi,np.pi,101)
X,Y = np.meshgrid(x,y)
Z = 3*X*np.log(np.sqrt(X**2+Y**2)) - np.cos(X*Y**2 - np.pi*X)
ax1 = fig.add_subplot(121)
clb = ax1.contourf(X,Y,Z, 31, cmap=cm.coolwarm)
ax1.set_xlabel('x')
ax1.set_ylabel('y')
ax1.grid()
fig.colorbar(clb, ax=ax1)

x = np.linspace(-3,3,7)
y = np.linspace(-np.pi,np.pi,7)
X,Y = np.meshgrid(x,y)
ax2 = fig.add_subplot(122)
U = 3*np.log(np.sqrt(X**2+Y**2)) + 3*X*X/(X**2+Y**2) + np.sin(X*Y**2 - np.pi*X)*(Y*
V = 3*X*Y/(X**2+Y**2) + 2*Y*np.sin(X*Y**2 - np.pi*X)*(2*X*Y)
```

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ax2.quiver(X,Y,U,V,scale_units='xy', angles='xy', scale=10)
ax2.set_xlabel('x')
ax2.set_ylabel('y')
ax2.grid()
fig.suptitle('3')
plt.show()
gx = 3*np.log(np.sqrt(1**2+0**2)) + 3*1/(1**2+0**2) + np.sin(1*0**2 - np.pi*1)*(0
gy = 3*1*0/(1**2+0**2) + np.sin(1*0**2 - np.pi*1)*(2*1*0)
print('Gradient an [1,0]: x = {0:.4f}, y= {1:.4f}'.format(gx,gy))
```

3)



Gradient an [1,0]: x = 3.0000, y = 0.0000

Partielle Ableitungen höherer Ordnung

Gegeben ist die Funktion $z(x, y) = 3xy - \cos(x - y) + x^3y^5$.

a) Zeigen Sie: $z_{xy} = z_{yx}$

b) Zeigen Sie: $z_{xxy} = z_{xyx} = z_{yxx}$

Hinweis zur Notation: $f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x \partial y}$, $f_{xx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2}$

Lösung

a)

$$z_x = 3y + \sin(x - y) + 3x^2y^5$$

$$z_y = 3x - \sin(x - y) + x^35y^4$$

$$z_{xy} = \frac{\partial}{\partial y}(z_x) = 3 - \cos(x - y) + 3x^25y^4$$

$$z_{yx} = \frac{\partial}{\partial x}(z_y) = 3 - \cos(x - y) + 3x^25y^4$$

b)

$$z_{xx} = \frac{\partial}{\partial x}(z_x) = \cos(x - y) + 6xy^5$$

$$z_{xxy} = \frac{\partial}{\partial y}(z_{xx}) = \sin(x - y) + 6x5y^4$$

$$z_{xyx} = \frac{\partial}{\partial x}(z_{xy}) = \sin(x - y) + 6x5y^4$$

$$z_{yxx} = \frac{\partial}{\partial x}(z_{yx}) = \sin(x - y) + 6x5y^4$$

Partielle Ableitungen höherer Ordnung

Zeigen Sie, dass

$$u(x, y, z) = \frac{a}{\sqrt{x^2 + y^2 + z^2}} + b$$

eine Lösung der folgenden partiellen Differentialgleichung (Laplace-Gleichung)

$$u_{xx} + u_{yy} + u_{zz} = 0$$

ist.

Lösung

Struktur der partiellen Ableitung 1. Ordnung:

$$u_x = -\frac{ax}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}$$

- analoge Struktur für die anderen partiellen Ableitungen 1. Ordnung

partielle Ableitungen 2. Ordnung:

$$u_{xx} = \frac{a(2x^2 - y^2 - z^2)}{(x^2 + y^2 + z^2)^{\frac{5}{2}}}$$

$$u_{yy} = \frac{a(2y^2 - x^2 - z^2)}{(x^2 + y^2 + z^2)^{\frac{5}{2}}}$$

$$u_{zz} = \frac{a(2z^2 - x^2 - y^2)}{(x^2 + y^2 + z^2)^{\frac{5}{2}}}$$

Einsetzen in Laplace-Gleichung:

$$\frac{a(2x^2 - y^2 - z^2) + a(2y^2 - x^2 - z^2) + a(2z^2 - x^2 - y^2)}{(x^2 + y^2 + z^2)^{\frac{5}{2}}} = \frac{a(2x^2 - x^2 - x^2 + 2y^2 - y^2 - y^2 + 2z^2 - z^2 - z^2)}{(x^2 + y^2 + z^2)^{\frac{5}{2}}} = 0$$

```
In [3]: # wertebereich der parameter
x = np.linspace(1,3,11)
y = np.linspace(1,3,11)
z = np.linspace(1,3,11)

# parameter der funktion
a = 1
b = 1

fig, axs = plt.subplots(1,3,figsize=(15,4))

# in der x-y-Ebene
cax = axs[0]
xx, yy = np.meshgrid(x,y)
zz = 0
u = a*(2*xx**2 - yy**2 - zz**2)/(xx**2 + yy**2 + zz**2)**(5/2)
v = a*(2*yy**2 - xx**2 - zz**2)/(xx**2 + yy**2 + zz**2)**(5/2)
w = a*(2*zz**2 - xx**2 - yy**2)/(xx**2 + yy**2 + zz**2)**(5/2)
cax.contourf(xx, yy, a/(xx**2 + yy**2 + zz**2)**(1/2) + b, 20, alpha=0.6)
cax.quiver(xx, yy, u, v, scale=0.75, scale_units='xy', angles='xy', color='red')
cax.set_xlabel('x')
cax.set_ylabel('y')
```

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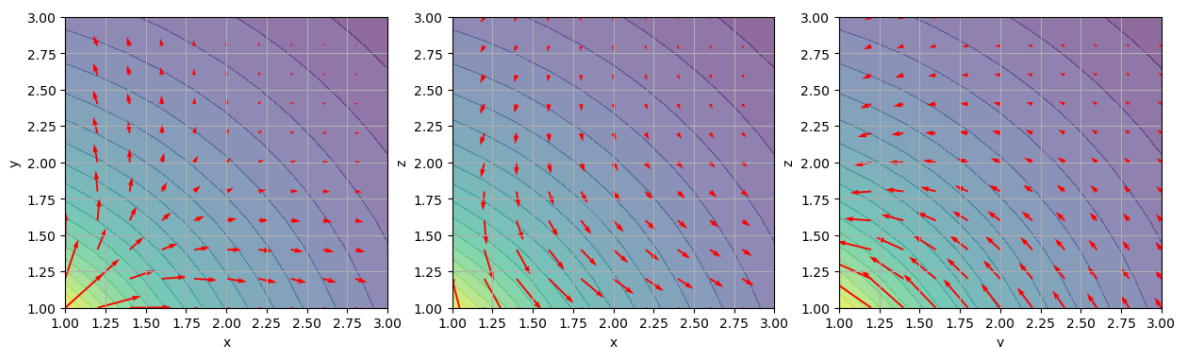
cax.grid()

# in der x-z-Ebene
cax = axs[1]
xx, zz = np.meshgrid(x,z)
yy = 0
u = a*(2*xx**2 - yy**2 - zz**2)/(xx**2 + yy**2 + zz**2)**(5/2)
v = a*(2*yy**2 - xx**2 - zz**2)/(xx**2 + yy**2 + zz**2)**(5/2)
w = a*(2*zz**2 - xx**2 - yy**2)/(xx**2 + yy**2 + zz**2)**(5/2)
cax.contourf(xx, zz, a/(xx**2 + yy**2 + zz**2)**(1/2) + b, 20, alpha=0.6)
cax.quiver(xx, zz, u, v, scale=0.75, scale_units='xy', angles='xy', color='red')
cax.set_xlabel('x')
cax.set_ylabel('z')
cax.grid()

# in der y-z-Ebene
cax = axs[2]
yy, zz = np.meshgrid(y,z)
xx = 0
u = a*(2*xx**2 - yy**2 - zz**2)/(xx**2 + yy**2 + zz**2)**(5/2)
v = a*(2*yy**2 - xx**2 - zz**2)/(xx**2 + yy**2 + zz**2)**(5/2)
w = a*(2*zz**2 - xx**2 - yy**2)/(xx**2 + yy**2 + zz**2)**(5/2)
cax.contourf(yy, zz, a/(xx**2 + yy**2 + zz**2)**(1/2) + b, 20, alpha=0.6)
cax.quiver(yy, zz, u, v, scale=0.75, scale_units='xy', angles='xy', color='red')
cax.set_xlabel('y')
cax.set_ylabel('z')
cax.grid()

plt.show()

```



Totales Differential

Bestimmen Sie das totale Differential folgender Funktionen:

- $z(x, t) = \frac{t^2 + x}{2t - 4x}$
- $z(u, v, w) = \ln \sqrt{u^2 + v^2 + w^2}$
- $z(x, y) = 4x^3y - 3e^yx$

Lösung

Totales Differential: $d\varphi = \frac{\partial \varphi}{\partial x} dx + \frac{\partial \varphi}{\partial y} dy + \frac{\partial \varphi}{\partial z} dz$

- $dz = \frac{2t^2 - 8tx - 2x}{(2t - 4x)^2} dt + \frac{4t^2 + 2t}{(2t - 4x)^2} dx$

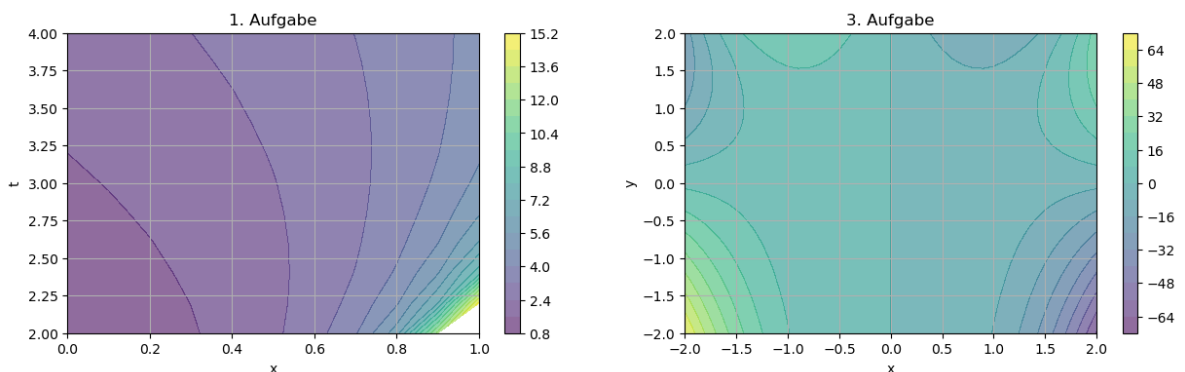
- $dz = \frac{u}{u^2 + v^2 + w^2} du + \frac{v}{u^2 + v^2 + w^2} dv + \frac{w}{u^2 + v^2 + w^2} dw$
- $dz = (12x^2y - 3e^y)dx + (4x^3 - 3xe^y)dy$

```
In [4]: fig, axs = plt.subplots(1,2, figsize=(15,4))
cax = axs[0]
x = np.linspace(0,1,11)
t = np.linspace(2,4,11)
xx, tt = np.meshgrid(x,t)
clb = cax.contourf(xx, tt, (tt**2+xx)/(2*tt-4*xx), 20, alpha=0.6)
cax.set_xlabel('x')
cax.set_ylabel('t')
cax.set_title('1. Aufgabe')
cax.grid()
fig.colorbar(clb, ax=cax)

cax = axs[1]
x = np.linspace(-2,2,41)
y = np.linspace(-2,2,41)
xx, yy = np.meshgrid(x,y)

clb = cax.contourf(xx, yy, 4*xx**3*yy - 3*np.exp(yy)*xx, 20, alpha=0.6)
cax.set_xlabel('x')
cax.set_ylabel('y')
cax.set_title('3. Aufgabe')
fig.colorbar(clb, ax=cax)
cax.grid()

plt.show()
```



Kettenregel

Berechnen Sie unter Anwendung der Kettenregel $\dot{z} = dz/dt$ für

$$z(x, y) = x^2 + 6xy + 2y^2,$$

mit $x(t) = \cos(t)$, $y(t) = \sin(t)$.

Welchen Wert hat $\dot{z}$ für $t = \pi/2$? Wie sieht diese Funktion grafisch aus?

Lösung

Kettenregel: $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$

- $\frac{\partial z}{\partial x} = 2x + 6y$

- $\frac{dx}{dt} = -\sin(t)$
- $\frac{\partial z}{\partial y} = 6x + 4y$
- $\frac{dx}{dt} = \cos(t)$

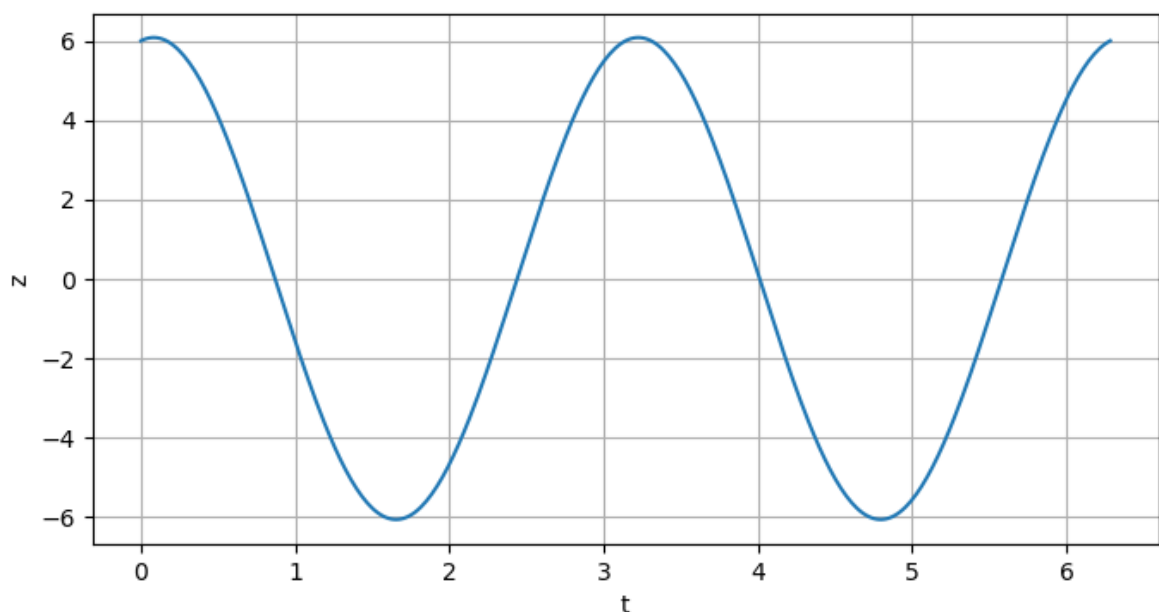
$$\frac{dz}{dt} = -(2x + 6y) \sin(t) + (6x + 4y) \cos(t) = -(2 \cos(t) + 6 \sin(t)) \sin(t) + (6 \cos(t) + 4 \sin(t)) \cos(t)$$

Anmerkung: $\cos^2(t) - \sin^2(t) = \cos(2t)$

```
In [5]: t = np.linspace(0, 2*np.pi, 361)
z = 2*np.cos(t)*np.sin(t) + 6*np.cos(2*t)

fig = plt.figure(figsize=(8,4))
ax = fig.add_subplot(111)
ax.plot(t,z)
ax.set_xlabel('t')
ax.set_ylabel('z')
ax.grid()
plt.show()

t = np.pi/2
print('z at pi/2: {:.4f}'.format(2*np.cos(t)*np.sin(t) + 6*np.cos(2*t)))
```



z at pi/2: -6.0000

Anwendung – Tangentialebene

Bestimmen Sie die Gleichung der Tangentialebene in $P=(0;1;0)$ für die Funktion

$$z(x, y) = 4x^3y - 3xe^y.$$

Korrektur: Der Punkt hatte eine falsche Z-Koordinate. $(0;1;1)$ liegt nicht auf der Oberfläche.

Lösung

$$z_x = 12x^2y - 3e^y$$

$$z_y = 4x^3 - 3xe^y$$

$$z_x(0, 1) = -3e$$

$$z_y(0, 1) = 0$$

Tangentialebene:

$$z - z_0 = z_x(x_0, y_0) * (x - x_0) + z_y(x_0, y_0) * (y - y_0)$$

$$z = -3e(x - 0) + 0 * (y - 1) = -3ex$$

Anwendung – Implizite Differentiation

Berechnen Sie die Tangentensteigung der Funktion $y(x)$ an dem Punkt, an dem $y = 3$ und $x < 0$, wenn diese Funktion implizit definiert ist: $\frac{x^2}{36} + \frac{y^2}{16} = 1$.

Lösung

$$y = 3 \wedge x < 0 \Rightarrow x^2 = 36 - y^2 \frac{36}{16} \Rightarrow x = \frac{3}{2}\sqrt{7}$$

$$F(x, y) = \frac{x^2}{36} + \frac{y^2}{16}$$

$$y'(x) = -\frac{F_x}{F_y} = -\frac{\frac{2x}{36}}{\frac{2y}{16}} = -\frac{4x}{9y}$$

$$y'(\frac{3}{2}\sqrt{7}) = -\frac{4\frac{3}{2}\sqrt{7}}{9*3} = -\frac{2}{9}\sqrt{7}$$

Optional

Zeigen Sie, dass die gemischten Ableitungen zweiter Ordnung der Funktion $f(x, y)$ bei $(x, y) = (0, 0)$ unterschiedlich sind.

$$f(x, y) = \begin{cases} \frac{xy^3 - x^3y}{x^2 + y^2} & \text{for } (x, y) \neq (0, 0) \\ 0 & \text{for } (x, y) = (0, 0) \end{cases}$$

(1) For $(x, y) \neq (0, 0)$ (using quotient rule): $f_x(x, y) = \frac{-x^4y - 4x^2y^3 + y^5}{(x^2 + y^2)^2}$

(2) For $(x, y) = (0, 0)$ (from definition):

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(0 + h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

(3) For $(x, y) \neq (0, 0)$ (using quotient rule): $f_y(x, y) = \frac{-x^5 + 4x^3y^2 + xy^4}{(x^2 + y^2)^2}$

(4) For $(x, y) = (0, 0)$ (from definition):

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, 0 + h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

(5) $f_{xy}(0, 0) = \lim_{h \rightarrow 0} \frac{f_x(0, 0 + h) - f_x(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{h^5}{h^4} - 0}{h} = 1$

(6) $f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(0 + h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{-h^5}{h^4} - 0}{h} = -1$

In []: